

## HOW TO USE

# KARBHARI

Working Capital Guardian — project guide

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Run it, give it a case, and read what it finds. Everything in this guide was captured from a live run of the real app on the test files that ship with the project.

**1**

Start the app

**2**

Open a case & attach  
evidence

**3**

Run the investigation

**4**

Read, verify, act

## STEP 1 — START THE APP

# Two ways to run it.

## One API key.

KARBHARI is one FastAPI backend serving a React console. Run it as a single Docker container, or run backend and frontend separately while developing. Either way you need a Google AI Studio (Gemini) API key; the free tier is enough.

### Before you start: the API key

Copy `backend/.env.example` to `backend/.env` and fill it in. Keep this file out of git.

```
GOOGLE_API_KEY=your-key-from-aistudio.google.com/apikey
LLM_PROVIDER=gemini
LLM_MODEL=gemini-flash-lite-latest # an alias, tracks Google's current model
```

## OPTION A • RECOMMENDED

#### Docker — the whole product in one container

```
# from the karbhari/ folder
docker build -t karbhari:0.5.0 .
run.bat
```

`run.bat` replaces any old `karbhari` container and starts the new one. Open <http://127.0.0.1:8000>.

## OPTION B • DEVELOPMENT

#### Backend + frontend, two terminals

```
# 1 • backend (Python 3.12+)
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
uvicorn app.main:app --port 8000

# 2 • frontend (Node 20+)
cd frontend
npm install
npm run dev
```

Open <http://127.0.0.1:5173>. Vite forwards `/api` to port 8000.

### Check it is alive

```
curl http://127.0.0.1:8000/api/health
→ {"status":"ok","service":"karbhari-backend"}
```

Data lives in `backend/data/` (SQLite database + uploaded files). Set `KARBHARI_DATA_DIR` to keep a separate workspace, for example a clean one for demos.

## STEP 2 – OPEN A CASE, ATTACH EVIDENCE

# One case per review.

## Tag every file.

**a Create the case.** On the home screen, enter a case name (e.g. *Q3 Drawing Power review*) and, optionally, the business name, then click **New case**. Open it from the list.

CASES – EVERY WORKING-CAPITAL INVESTIGATION STARTS HERE

**b Attach the evidence.** In the case, open **Evidence**, choose a file, pick its category, then click **Attach evidence**. Repeat for each document. PDF (with a text layer), XLSX, CSV and TXT all work.

| FILE                         | CATEGORY        | SIZE   | UPLOADED              |        |
|------------------------------|-----------------|--------|-----------------------|--------|
| sanction_letter.pdf          | sanction letter | 1.0 KB | 10/1/2026, 5:27:38 PM | Remove |
| stock_statement_june2026.txt | stock statement | 231 B  | 10/1/2026, 5:27:38 PM | Remove |
| stock_statement_sep2026.pdf  | stock statement | 963 B  | 10/1/2026, 5:27:38 PM | Remove |
| debtor_invoice_ledger.xlsx   | debtor ledger   | 5.0 KB | 10/1/2026, 5:27:39 PM | Remove |
| debtor_receipts.csv          | debtor ledger   | 224 B  | 10/1/2026, 5:27:39 PM | Remove |
| creditor_ledger.csv          | creditor ledger | 108 B  | 10/1/2026, 5:27:39 PM | Remove |

EVIDENCE – THE SIX TEST FILES FROM ASSETS FOR TESTING/, EACH TAGGED

| CATEGORY               | UPLOAD                                                        | GIVES KARBHARI                               |
|------------------------|---------------------------------------------------------------|----------------------------------------------|
| <b>sanction letter</b> | The bank's sanction / facility letter                         | Limit, margins, debtor age cut-off           |
| <b>stock statement</b> | Monthly stock statements (attach two months if you have them) | Stock value; period-over-period check        |
| <b>debtor ledger</b>   | Invoice-level ledger and receipts / payment records           | Eligible debtors, rebuilt invoice by invoice |
| <b>creditor ledger</b> | Sundry creditors list                                         | The creditor deduction                       |
| <b>bank statement</b>  | CC account statement showing the DP the bank recognises       | The figure to compare against                |

## STEP 3 — RUN THE INVESTIGATION

# Click once.

## Read the headline, then the findings.

Click **Run investigation** at the bottom of the left rail. The agent reads every document, cross-checks them and computes Drawing Power. On the test case it took **about 25–30 seconds**, in 10 steps.

The screenshot displays the KARBHARI Working Capital Guardian interface. On the left, a sidebar shows the 'Q3 Drawing Power review' for 'Suryoday Textiles Pvt Ltd'. It indicates a 'POTENTIAL CAPACITY IDENTIFIED' of ₹2.20L, which 'Appears supportable under the supplied facility terms, within the existing sanctioned limit.' The status is 'Complete' with 6 pieces of evidence and 3 findings. The main panel, titled 'FINDINGS', lists three items:

- SUPPORTED** (₹2,20,000): Calculated Drawing Power appears to exceed what the bank is currently recognising. Based on the sanction terms and the independently reconciled stock/debtor/creditor records, the calculated Drawing Power is 2570000.0, against a reported drawing power of 2350000.0. That is a gap of 220000.0 that appears supportable under the supplied facility terms, within the existing sanctioned limit -- not a request for more credit. This is not a guarantee the bank will recompute or release this amount.
- SUPPORTED** (₹6,00,000): Ineligible Book Debts Exceeding 90 Days. Debtor reconciliation identified invoices INV-1002 (Rs. 1,00,000 net balance) and INV-1005 (Rs. 5,00,000 net balance) as ineligible due to exceeding the 90-day aging limit stipulated in the sanction terms.
- SUPPORTED**: Stock and Book Debts Period-over-Period Swings. Significant period-over-period increases were observed between June 2026 and September 2026 across total stock (+25.0%), sundry debtors (+29.4%), and sundry creditors (+33.3%), exceeding the 20% threshold.

## FINDINGS — LIVE RESULT ON THE BUNDLED TEST CASE

### The left rail: your headline

- **Potential capacity identified** — the Drawing Power gap: here ₹2.20L (calculated ₹25,70,000 vs ₹23,50,000 recognised by the bank).
- Evidence count, findings count and run status.
- Tabs for every view, and **Run investigation** to re-run after adding files.

### Findings: how sure each one is

- **SUPPORTED** evidence and a tool output directly back it.
- **UNRESOLVED** incomplete or ambiguous, or an assumption was needed.
- **CONTRADICTED** the evidence works against the claim.
- Rupee impact top-right; **Supporting evidence** links to the exact documents.

The gap is headroom **within the existing sanctioned limit** that appears supportable under the facility terms. It is not a request for more credit, and not a guarantee the bank will release it.

STEP 4 – CHECK THE NUMBERS

# Every figure, shown in full.

Open **Investigation** to see the working behind the headline.

Status: complete · considered 6 evidence item(s) · started 10/1/2026, 5:29:01 PM

Investigation completed in 10 step(s). Calculated Drawing Power exceeds the reported figure by 220000.0. 3 finding(s) (3 supported, 0 unresolved, 0 contradicted/ineligible).

Drawing Power reconciliation

|               |                        |           |
|---------------|------------------------|-----------|
| Calculated DP | reported drawing_power | Gap       |
| ₹25,70,000    | ₹23,50,000             | ₹2,20,000 |

Debtor reconciliation

|                   |                      |
|-------------------|----------------------|
| Total outstanding | Eligible outstanding |
| ₹18,00,000        | ₹12,00,000           |

| INVOICE  | DATE       | AMOUNT    | BALANCE   | STATUS  | AGE (DAYS) | ELIGIBLE |
|----------|------------|-----------|-----------|---------|------------|----------|
| INV-1002 | 2026-05-01 | ₹3,00,000 | ₹1,00,000 | partial | 152        | no       |
| INV-1005 | 2026-06-15 | ₹6,00,000 | ₹5,00,000 | partial | 107        | no       |
| INV-1003 | 2026-07-20 | ₹7,00,000 | ₹0        | paid    | 72         | no       |
| INV-1001 | 2026-08-10 | ₹5,00,000 | ₹5,00,000 | unpaid  | 51         | yes      |
| INV-1004 | 2026-09-05 | ₹4,00,000 | ₹4,00,000 | unpaid  | 25         | yes      |
| INV-1006 | 2026-09-15 | ₹3,00,000 | ₹3,00,000 | unpaid  | 15         | yes      |

Period-over-period variance

total stock changed +25.0% from June 2026 (2400000.0) to September 2026 (3000000.0) -- worth confirming what drove this before relying on the latest figure alone.

sundry debtors changed +29.4% from June 2026 (1700000.0) to September 2026 (2200000.0) -- worth confirming what drove this before relying on the latest figure alone.

sundry creditors changed +33.3% from June 2026 (300000.0) to September 2026 (400000.0) -- worth confirming what drove this before relying on the latest figure alone.

INVESTIGATION – RECONCILIATION DETAIL FROM THE LIVE RUN

DRAWING POWER RECONCILIATION

Calculated DP, the bank's figure and the gap. Any assumption the engine had to make (a missing margin or creditor figure) is listed here; if inputs are missing, it names them.

DEBTOR RECONCILIATION

Every invoice with balance, status, age and eligibility. In the test case, INV-1002 (152 days) and INV-1005 (107 days) are past the 90-day cut-off, so **₹6,00,000** of the ₹18,00,000 outstanding is excluded. Duplicates and unallocated receipts appear here too.

PERIOD-OVER-PERIOD VARIANCE

Any figure that moved 20% or more between statements: here stock +25.0%, debtors +29.4%, creditors +33.3% from June to September. Confirm what drove them before relying on the latest month.

## STEP 5 — AUDIT THE REASONING

# See exactly how it got there.

Open **How it investigated** for the step-by-step trace of the live run: the agent's reasoning, the tool it called and a summary of the result. Expand **Raw tool call** to see the exact input and output.

The screenshot displays the KARBHARI interface. On the left, a sidebar shows the case 'Q3 Drawing Power review' for 'Suryoday Textiles Pvt Ltd'. The main panel, titled 'HOW IT INVESTIGATED', shows a sequence of five steps:

- 1 list\_evidence**: I need to begin the investigation by listing all available evidence items. 6 evidence item(s) listed. ▶ Raw tool call
- 2 read\_evidence**: I have listed the evidence. Step 2 requires me to read every single evidence item returned by list\_evidence. I will start by reading the first item: sanction\_letter.pdf. Read sanction\_letter.pdf (399 chars). ▶ Raw tool call
- 3 read\_evidence**: I have read the sanction letter. Now I must continue Step 2 by reading the remaining evidence items one by one: stock\_statement\_june2026.txt, stock\_statement\_sep2026.pdf, debtor\_invoice\_ledger.xlsx, debtor\_receipts.csv, and creditor\_ledger.csv. I will read the next item, stock\_statement\_june2026.txt. Read stock\_statement\_june2026.txt (229 chars). ▶ Raw tool call
- 4 read\_evidence**: I need to continue reading the remaining evidence items as mandated by Step 2. The next item to read is stock\_statement\_sep2026.pdf. Read stock\_statement\_sep2026.pdf (297 chars). ▶ Raw tool call
- 5 read\_evidence**: I have read the sanction letter and both stock statements. Now I need to continue reading the remaining evidence items: debtor\_invoice\_ledger.xlsx, debtor\_receipts.csv, and creditor\_ledger.csv, as required by Step 2. Read debtor\_invoice\_ledger.xlsx (378 chars). ▶ Raw tool call

HOW IT INVESTIGATED — THE REAL TRACE, RECORDED AS IT RAN

## WHAT A COMPLETE RUN LOOKS LIKE

list\_evidence → read\_evidence × every file →  
check\_consistency → reconcile\_debtors →  
calculate\_drawing\_power → finish. Up to 20  
steps.

## ACTIONS TAB

Reserved for the next phase: the evidence pack and recommended corrective actions, such as a note to the bank. For now, take the findings and the Investigation tab to your banker or CA.

## TIPS &amp; TROUBLESHOOTING

# Getting a complete answer.

| IF YOU SEE...                                          | DO THIS                                                                                                                                                                                       |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNRESOLVED</b> Drawing Power was not calculated     | The agent finished before running the calculation. It is non-deterministic; in our test, one run of two did this. Click <b>Run investigation</b> again; each run is kept in the case history. |
| ...could not be independently calculated. Missing: ... | Upload the document named in the message: usually the sanction letter, a current stock statement or a debtor ledger.                                                                          |
| ...nothing to compare it against                       | Add a bank / CC statement (or a document stating the DP the branch recognises).                                                                                                               |
| Placeholder margins / creditors treated as zero        | The sanction letter or creditor ledger wasn't found or readable. The result is an estimate until you add them.                                                                                |
| Consistency was not checked in this run                | Two same-category statements exist but the agent skipped the comparison. Re-run, or compare the periods yourself.                                                                             |
| "No extractable text found"                            | The PDF is a scan. There is no OCR yet; upload a text PDF, Excel or CSV export instead.                                                                                                       |
| Investigation could not start                          | Check <code>GOOGLE_API_KEY</code> in <code>backend/.env</code> and restart.                                                                                                                   |
| Port 8000 already in use                               | Another KARBHARI (often the Docker container) is running. Stop it, or start uvicorn with <code>--port 8010</code> .                                                                           |
| Code change not showing (Windows)                      | <code>--reload</code> can miss changes. Stop all Python processes and start fresh.                                                                                                            |

## FOR THE BEST RESULT, UPLOAD

- The sanction letter, with margins and the debtor age limit
- Two consecutive stock statements
- Invoice-level debtors *and* receipts, not just a subtotal
- A creditor list and the bank's recognised DP

## TRY IT NOW

The six files in `karbhari/ASSETS FOR TESTING/` reproduce the run in this guide: a **₹2.20L** gap and 3 supported findings.

Run the tests: `cd backend` then `python -m pytest -q`.